

Environmental Spill Procedure

PURPOSE

The purpose of this procedure is to provide guidance and assistance with the development and production of specific procedures with regards to environmental spillages. It is the responsibility of all personnel to prevent spills through appropriate material storage and handling practices but should these controls fail, this procedure is needed to provide guidance on:

SCOPE

This procedure applies to all employees and operational activities. The responsibilities for implementing the procedure are outlined below.

Site/Location Manager

- Overall responsibility for prevention and control of potential spillages on site
- Develop section of the Environmental Management Plan to include prevention and control spillages
- Liaise with the Environmental Manager.

Environmental Manager

- Assist Site Managers in the development of the Environmental Management Plan regarding the prevention and control of spillage issues
- Ensure all staff comply with the guidelines

PROCEDURES

Once the safety risks of the spilt product have been assessed, the following guidance to mitigate the spill should be considered along with the specific circumstances of the spill, before deciding how best to stop, contain and recover the spillage.

Stop the Leak at the Source

Following assessment of safety risks of the material, consider actions required to control the source of the spill this will include:

Contain the Spill

Identify proximity of sensitive receptors including the workforce, the surrounding community or watercourses. Also consider the location of ignition sources or incompatible chemicals that may increase the risks presented by the spill.

Contain the spill by using absorbent material from the appropriate spill kit

Oil and Fuel Spill Kits – are to be used for diesel, fuel, unleaded petrol, hydraulic oils, grease and other petroleum based lubricant spills. They include oil absorbent booms, pillows and pads.

Segregate Contaminated Material

Ensure the contaminated area is segregated with the installation of appropriate demarcation barriers.

In the event of a significant spill, it may be prudent to relocate the affected material to a identified exclusion zone.

Some options for containing a spill include:

- a. Creating an earthen perimeter bund or using oil absorbent booms to act as a barrier;
 - b. Preventing access into stormwater drains by:
 - i. Using drain covers over pits (drain covers found in all small oil spill kits.;
 - ii. Plugging the stormwater pipe using inflatable fenders;
 - c. Use of absorbent pads around the edge of the hydrocarbon spill;
 - d. Use of hydrophobic booms to restrict the flow of liquid and keep it pooled within the booms;
 - e. Redirecting the flow of material to an area that poses less safety or environmental risk;
 - f. Identify any other reasonable actions that may improve containment.
- As a last resort, other materials such as sand may be used. Avoid using these materials as they create a large amount of contaminated waste.
 - For a solids spill, dust suppression measures may be required if dust is an issue.

Clean-up the Spill

Once the source of the spill has been controlled and the spill contained (immediate threat to the environment has been minimised) the rest of the spill will need to be cleaned-up.

- oil/fuel spills, continue to use absorbent materials from oil spill kits around site. Used and contaminated absorbent materials should be collected in either 220L drums or in lined skip bins, depending on the size of the clean-up. This applies for spills on sealed and gravel areas
- For mineral concentrate spills scrape up the spillage, use contractor to suck up the product or wash down the affected area. Do not let wash waters enter stormwater drains and contact the Environmental Team to request the most applicable disposal method. Where possible, the material shall be recovered and recycled. This may apply to:
 - a. ore concentrate
 - b. sulphur granules
 - c. chemicals

d. waste oil / grease

If material cannot be recycled, advice should be sought from the Environment Department on appropriate disposal methods to ensure compliance with the Environment Protection Act

Restock Kits

Any personnel who use spill response equipment must ensure it is appropriately replenished and returned to its storage position prior to resuming work.

Reporting

Staff member who was notified of the spill is to complete an incident report with the Environment Team

Spill Hierarchy

The spill hierarchy depends on the risks of the spilt material that depends on the type of substance spilt as well as the volume of the spill. The following presents guidance on describing a category for the spill.

Minor Spills

A minor spill is typically:

- easily contained and so does not pose a risk to human health or the environment;
- can be cleaned up by an individual or within the work area using spill response equipment;
- Providing there is total clean-up of a spill on a sealed area it can be reported as a near miss; and
- If the spill occurs on an unsealed area, it is an emission to land and groundwater and must be reported as an incident.

Oil contamination of unsealed areas should be avoided. Spillage on an unsealed area is an environmental incident as it may contaminate soil, groundwater and stormwater. Spillage on a sealed area should be bunded, recovered and restricted to a temporary hazard.

Significant Spills

A significant spill may:

- Not be readily contained by an individual or within the work area using the available spill response equipment;
- May require the involvement of the site's team members from other work areas.
- Likely to endanger safety, result in pollution or disrupt operations;
- Is an incident and requires reporting and investigation by the Environment Team.

Major Spills

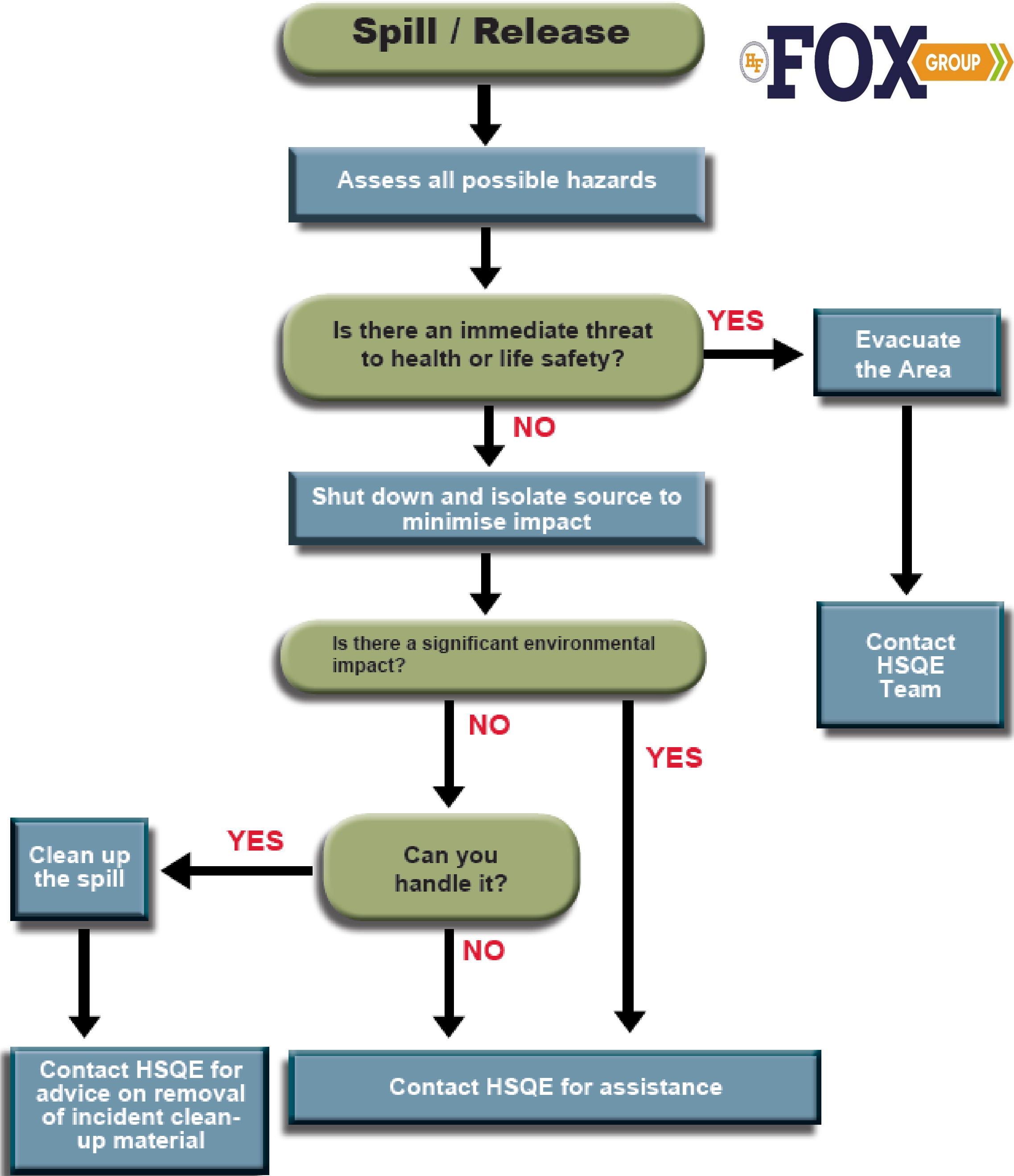
A major spill typically requires:

- Evacuation of personnel and the immediate involvement of the Emergency Services.

Examples of major spills are:

- Oil spill into stormwater system or directly into the ocean;
- Flammable materials that are uncontained;
- Hazardous gases that are filling a confined space

Spill Response Procedure



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